TECHNICAL OVERVIEW - V5.1 ARCHITECTURE

Chama

A non-custodial, peer-to-peer Bitcoin marketplace for local money - built on Nostr, Fedimint, Lightning, and verifiable on-chain commitment.

> Chama coordinates trust. Trade money exists only for the lock-claim window. A bond is different: an owner's own Bitcoin, made deliberately illiquid in public, for a term.

Seller

locks the sats

Buyer

claims the sats

Arbiter

breaks ties

One client composes four independent trust systems

Chama has no central account server and never becomes the custodian. It coordinates primitives that already know how to prove identity, move value, issue ecash, and enforce time.

Nostr - coordination

Nostr keys are identity. Signed events carry listings, encrypted trade state, chat, rosters, ratings, premiums, and bond announcements. Any client can replay the event history.

Fedimint - trade escrow

Ecash is minted for an active trade, split into holder-only Shamir shares, reconstructed after 2-of-3 agreement, and redeemed. Browser clients use WASM; native clients use the Rust bridge.

Lightning - value interface

Lightning moves sats into and out of the trade. It bridges the user's wallet and the selected federation without making Chama the money endpoint.

Bitcoin - commitment clock

Arbiter bonds are Taproot outputs locked by Bitcoin consensus until a block height. They are not Fedimint escrow and are not held by another person.

Architecture boundary: trade escrow answers "who receives these sats?" The commitment bond answers "has this person made their own capital illiquid, publicly, for long enough to earn durable privileges?"

The Woven Trust model

Every funded trade has three holders and a threshold of two. No single participant - including the arbiter - can move the ecash alone.

Seller + Buyer + Arbiter

three encrypted shares - any two reconstruct - one never can

Happy path

Buyer and seller agree. Their two shares reconstruct the ecash. The arbiter remains outside the money path.

Dispute

The arbiter joins one party to form the threshold. Roles derive from signing keys; votes are immutable; valid recipients are constrained by the trade.

Trade lifecycle

BOLT11 **fund** › mint + split **LOCK** › 2-of-3 **VOTE** › reconstruct **CLAIM** › Lightning **payout**

Funding and locking are one atomic effect. The reducer is deterministic and replayable. Crash-safe journals and reconciliation prevent a lost response from becoming a second payment.

Same protocol, three surfaces

Web

React + Vite with the Fedimint WASM client.

Android

Capacitor shell packages the web UI and native Rust Fedimint bridge.

Desktop

Tauri shell runs the same UI and bridge as a sidecar.

Code boundaries

LAYER	RESPONSIBILITY
`escrow-engine`	Event validation, state transitions, replay, relay coordination.
`fedimint`	Mint operations, recovery, payout, browser/native adapters.
`bond-multisig`	Taproot CLTV bond construction, funding detection, reclaim, announcements.
`arbiters`	Roster authority, bonded seating, exposure, liveness, premiums.
`ui`	Platform-neutral product language and interaction.

Nostr kinds: the trade family currently spans `38100–38113`, with retired numbers reserved. Roster, reputation, and chain-backed bond announcements live in separate allocations, including `38135`.

Lock your own sats to your own key

The shipped bond is a public time commitment, not slashable custody.

<T> CLTV DROP <owner key> CHECKSIG

1	0	T
owner	custodians	unlock height

- The owner funds a Taproot address derived from their x-only bond key and an absolute block height.
- The internal key is a BIP341 NUMS point, so the key path is unspendable. Every reclaim must satisfy the visible timelock leaf.
- Before height T, Bitcoin consensus rejects the spend. After T, only the owner signs the reclaim.
- The signal is **how much x how long**: capital made illiquid in public.

Nobody can seize it. Nobody can return it. Nobody can collude over it. The arbiter's own inability to withdraw early is the commitment.

Why the old design had to die

The earlier design reused the trade's 2-of-3 Shamir mechanism for long-lived bonds: an owner held one share and two cabinet members held the others. That sounded elegant because it reused audited machinery. It was wrong because a standing cabinet became a custody and collusion surface.

Old: cabinet custody

- Long-lived Fedimint ecash
- Two outside custodians
- Availability risk at term end
- Self-dealing and collusion incentives
- "Slashing" by withholding recovery

Shipped: consensus commitment

- Native on-chain Bitcoin
- No outside key holder
- Deterministic CLTV maturity
- Owner-only reclaim
- Publicly verifiable funding and term

The fix did not add a better committee. It removed the committee.

A Nostr announcement is advisory, not trusted. Clients recompute the bond address from the owner key and lock height, query the chain, measure the actual confirmed sats, and compare the current tip with the maturity height.

Commitment becomes useful without becoming custody

Arbiter seating

Chain-verified active bonds make arbiters eligible for bonded seating. Unbacked declarations do not count as capital.

Community liveness

Clients combine bonded coverage, remaining commitment, and reputation into a legible signal of whether a chama can support real commerce.

Insurance earnings

A bonded arbiter can earn the settlement premium for covering a trade. The bond is an earnings license, not a pool Chama controls.

Recovery

After maturity the owner can reclaim to an external Bitcoin address, the bond key, or a federation deposit route, with safe fallback and durable tracking.

Separation of powers: Nostr proves who announced. Bitcoin proves what is funded and when it unlocks. Reputation records behavior. The escrow threshold still decides each trade.

A store persists; a trade still expires

Store permanence is shipped without extending the lifetime of locked money.

store listing **CREATE**	›	24h window **unfunded**	›	re-publish **renew**	›	same terms **fresh CREATE**

A listing's expiry also anchors the safety timeout once a trade locks. Making that expiry arbitrarily long would make storefronts look persistent by leaving buyer or seller funds exposed for too long. Chama instead renews the public offer by publishing a fresh listing only when the prior one was seller-owned and never funded.

SELLER	STORE BEHAVIOR	TRADE BEHAVIOR
Unbonded	24-hour listing; manual renewal from the seller's dashboard.	Normal short lock timeout.
Chain-verified bonded	Seven-day rolling tenure; automatic renewal while the seller is online.	The same normal short lock timeout.

Permanence by renewal, never permanence by longer locks. Renewal moves no sats and never resurrects a funded, settled, child, or deliberately cancelled trade.

The bond-store flywheel

The bond makes persistence costly to fake. Persistence makes the bond economically productive.

Bond capital x time › **Privileges** seat + tenure › **Presence** persistent store › **History** trades + ratings

For a buyer

A persistent storefront is attached to a Nostr identity whose active bond can be independently verified. It is not proof of honesty, but it is proof of costly continuity.

For a seller

The same bond that can license arbiter earnings also grants durable storefront tenure. Locked capital gains productive utility without being transferred to Chama.

The design resists two cheap attacks at once: disposable stores that endlessly respawn without history, and theatrical bond declarations with no coins behind them. The chain verifies the commitment; the relay carries the storefront; completed trades build reputation.

What the bond does not buy: control over another user's funds, a positive rating, immunity from dispute, or a longer trade timeout.

The foundation survives the new capabilities

Never lose a sat

Pending operations are durable. Unknown outcomes are reconciled, not blindly retried. Recovery paths prefer proof over cleanup.

Never send one twice

Intent is recorded before money-moving effects. Payout journals and idempotent recovery close crash windows.

No unilateral trade movement

Holder-only shares and 2-of-3 resolution remain the trade's consensus boundary.

No bond custodian

The bond key belongs to the owner. CLTV constrains time; no Chama actor gains a seizure key.

No immortal lock

Store renewal changes discoverability, never the timeout of an individual funded trade.

No trusted announcement

Bond addresses and amounts are recomputed and checked against Bitcoin before privileges are granted.

> Chama's architecture now has three durations: relay history can be long-lived; a storefront can be renewable; trade ecash remains short-lived. The commitment bond is long-lived only because Bitcoin itself enforces the exit date.

Current architecture: Nostr event coordination, Fedimint ecash escrow, Lightning funding and payout, native Rust bridges, chain-verified commitment bonds, bonded arbiter premiums, and bond-gated store persistence. Future covenant work may strengthen enforcement, but it is not required for the guarantees described here.